SWA-PTIS: Socially-Weighted Alignment with Prospect-Theory Importance Sampling for Cross-Cultural Moral Alignment of Large Language Models

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Abstract

Large language models reason about moral dilemmas through a single, largely Western lens, and closing this gap usually means fine-tuning, reward modelling, or per-culture labelled data—resources that do not exist at the scale of the world's cultures. We ask whether a frozen model can instead be steered toward country-specific moral preferences at inference time, using only public survey data. We propose **SWA-PTIS** (Socially-Weighted Alignment with Prospect-Theory Importance Sampling): for each target country we construct $N{=}4$ persona agents from World Values Survey microdata, extract their decision logits in a single batched forward pass, linearly debias the logits via a two-pass A/B swap, and apply one σ-normalised Prospect-Theory importance-sampling correction to the inter-persona disagreement. The method is training-free, one-shot, and adds a single batched forward pass at inference. On MultiTP across 15 countries and 12 native languages, SWA-PTIS reduces Jensen–Shannon distance to country-level human preferences by 37.6% on Qwen2.5-72B and 34.7% on Llama-3.1-70B, exceeding every prompting and activation-steering baseline we tried, while smaller 4-bit-quantised models show consistent directional gains. Mechanistic ablations isolate the Prospect-Theory gain/loss asymmetry as the load-bearing ingredient. Code: `[repository]`.

1 Introduction

When a self-driving car must choose between sparing a group of elderly men or a group of young women, there is no culturally universal answer. The Moral Machine experiment [Awad et al., 2018]—40 million judgments from 233 countries—revealed that such preferences vary profoundly across cultures, and the multilingual successor benchmark MultiTP [Jin et al., 2025] documents that none of 19 contemporary LLMs align broadly with non-WEIRD human preferences [Henrich et al., 2010]. LLMs deployed worldwide make moral decisions through one culturally monolithic lens inherited from their training data, and the gap does not close on its own.

Why the usual tools do not fit. Fine-tuning [Bai et al., 2022, Ouyang et al., 2022] requires per-culture labelled data and modifies weights irreversibly. Prompting-based steering is ad hoc [Jin et al., 2025]. Activation steering [Arditi et al., 2025, Zou et al., 2023] requires white-box access and per-model engineering. Reward-guided decoding [Khanov et al., 2024, Mudgal et al., 2024] needs trained reward models unavailable for cross-cultural settings. Culture-aware adapters [Li et al., 2024, Yao et al., 2024] scale poorly across 100+ countries. None of these methods can adapt a frozen LLM's moral reasoning to a target culture *at inference time* without labelled data or weight modification.

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Disagreement is signal. Moral preferences within any society are not monolithic—age cohorts and social strata exhibit systematically different priorities [Awad et al., 2018, Kleiman-Weiner et al., 2017]. The World Values Survey [Haerpfer et al., 2022] provides empirically calibrated data on these within-country differences for 80+ nations. Our insight is simple: when culturally grounded persona agents derived from WVS data *disagree* on a moral decision, that disagreement reveals that the decision is culturally contested, and the direction of disagreement can be turned into a bounded logit correction without any new labels.

SWA-PTIS in one paragraph. For each country c we construct $N{=}4$ persona agents from WVS microdata (three age cohorts plus a country-invariant utilitarian agent). We score every scenario by (i) running a single batched forward pass that returns the base-model decision logits plus one set of logits per persona, (ii) *linearly* debiasing those logits with a two-pass A$\leftrightarrow$B swap—the swap cancels an additive positional bias exactly at the logit level, before any nonlinearity—and (iii) running one Prospect-Theory importance-sampling correction on the debiased logits. The Prospect-Theory value function is fed a σ-normalised per-persona gain, so the utility is dimensionless and cross-dimensionally comparable; the softmax weights minimise a distributional KL to the Gaussian proposal by construction, and a K_{eff} rejection rule falls back to the persona consensus when the importance-weight estimator collapses.

Contributions.

- A training-free, one-shot inference-time steering method (§3) that turns within-country WVS variation into a bounded logit correction. The method adds one extra batched forward pass at inference and no new parameters.
- A clean mathematical formulation (§3.3): σ-normalised gain fed to Prospect Theory, mean-of-v aggregation that preserves loss aversion under Jensen's inequality, linear logit-level debiasing applied before the nonlinear importance-sampling step, and a K_{eff} safety rail.
- Empirical evidence on five models (7B–72B; §5). On the two strong instruction-tuned 70B-class backbones SWA-PTIS cuts mean Jensen–Shannon distance by 37.6% (Qwen2.5-72B) and 34.7% (Llama-3.1-70B) over vanilla, beating every prompting and activation-steering baseline we tried.
- A mechanism analysis (§5.3). Deterministic consensus-shift baselines and a Binary-ARGS variant that share SWA-PTIS's infrastructure pin the Prospect-Theory gain/loss asymmetry—not the WVS weighting alone—as the active ingredient.

2 Related Work

Cross-cultural moral benchmarks. The Moral Machine experiment [Awad et al., 2018] provides 40M+ cross-cultural moral judgments. MultiTP [Jin et al., 2025] extends it to 107 languages with country-specific preference rates; we adopt it as our evaluation and propose an alignment *intervention* rather than diagnostics alone. Other benchmarks (ETHICS [Hendrycks et al., 2021], MoralExceptQA [Jin et al., 2022], MoCa [Nie et al., 2023]) remain English-centric. PRISM [Kirk et al., 2024] and GlobalOpinionQA [Durmus et al., 2023] address diversity but cover few languages.

Inference-time alignment. Activation steering [Arditi et al., 2025, Zou et al., 2023] requires white-box access. ARGS [Khanov et al., 2024] and controlled decoding [Mudgal et al., 2024] need trained reward models unavailable for cross-cultural targets. CultureManager [Li et al., 2024] and CAReDiO [Yao et al., 2024] train culture-specific adapters. To the best of our knowledge, SWA-PTIS is the first training-free method targeting country-level moral calibration specifically. Sorensen et al. [2024] articulate the goal of pluralistic alignment, which SWA-PTIS operationalises concretely.

Prospect Theory and importance sampling. We use Kahneman–Tversky Prospect Theory [Kahneman and Tversky, 1979] as the loss-averse utility and wrap it in a self-normalised importance sampler with a Boltzmann softmax target. This is a standard variational construction [**?**], *not* path-integral MPPI [**?**]: our problem is scalar and one-step, which is why we call it Prospect-Theory Importance Sampling (PTIS).

3 Method: SWA-PTIS

3.1 Problem setup and notation

Let $\mathcal{X}$ be the set of moral dilemma scenarios, each presenting a forced binary choice via decision tokens `A` and `B`. For target country c, let $\mathbf{h}^{(c)} \in \mathbb{R}^6$ be the human preference-rate vector from Moral Machine, covering six moral dimensions (Species, Gender, Age, Fitness, Social Value, Utilitarianism). A frozen LLM f_θ produces logits $\mathbf{z}(\mathbf{x}) = [z_a, z_b]$ at the decision-token position, and the spare probability is $p_{\text{spare}} = \sigma(\Delta/T_{\text{dec}})$ with $\Delta = z_b - z_a$. Following the MultiTP convention [Jin et al., 2025], each dimension's model score is estimated as the empirical mean of p_{spare} on the scenarios of that dimension (a "mean preference rate"; Utilitarianism filters scenarios with n_{diff}=0). We report both the L2 misalignment score $\mathrm{MIS}(\mathbf{h}, \mathbf{m}) = \|\mathbf{h} - \mathbf{m}\|_2 \in [0, \sqrt{6}]$ [Jin et al., 2025] and Jensen–Shannon shape distance [Lin, 1991]; MIS is primary and JSD is a complementary shape diagnostic (Appendix A). **Goal:** without modifying θ, find a logit correction $\delta^\star(\mathbf{x}, c)$ such that the corrected preference vector is closer to $\mathbf{h}^{(c)}$.

3.2 Persona construction from the World Values Survey

For each country c we build $N = 4$ persona agents from WVS Wave 7 individual-level micro-data [Haerpfer et al., 2022]. Three agents correspond to age cohorts (young 18–35, middle 36–55, older >55), and the fourth is a country-invariant utilitarian anchor ("maximise lives saved"). For each age cohort we extract eight value dimensions—gender egalitarianism, religious importance, interpersonal trust, moral permissiveness, work centrality, family importance, personal autonomy, meritocratic orientation—and convert each cohort's mean into natural-language descriptors via fixed thresholds (Appendix B). Each persona becomes a system prompt of the form:

> *You are a [age-cohort] adult from [country]. The cultural values typical of your social group are: [WVS statistics]. When making moral judgments, you tend to [description reflecting WVS profile].*

This WVS grounding is the load-bearing design choice: the importance-sampling solver only decides *how* to combine persona disagreement, it does not invent cultural content. If the personas are wrong, SWA-PTIS will follow them; the persona construction pipeline is therefore the real foundation of the method.

3.3 Logit-level debiasing, then one Prospect-Theory IS step

Batched evaluation. All $N{+}1 = 5$ input sequences (base + 4 personas) are evaluated in a *single* batched forward pass:

$$\mathbf{z}_{\text{base}},\, \mathbf{z}_1, \ldots, \mathbf{z}_N \;=\; f_\theta(\mathbf{x};\, \varnothing,\, \mathbf{s}_1, \ldots, \mathbf{s}_N). \tag{1}$$

Per-category logit temperatures T_{cat} (Table 1) scale the logits before computing decision gaps: $\delta_i = (z_{i,b} - z_{i,a})/T_{\text{cat}}$ and $\delta_{\text{base}} = (z_{\text{base},b} - z_{\text{base},a})/T_{\text{cat}}$.

Two-pass linear debiasing. Each scenario is evaluated twice: once in the original ordering and once with choice labels and group labels swapped in the native language. We collect the logit gaps $(\delta^{(1)}_{\text{base}}, \delta^{(1)}_i)$ and $(\delta^{(2)}_{\text{base}}, \delta^{(2)}_i)$ from the two passes and combine them linearly *before* any nonlinearity:

$$\delta^{\text{deb}}_{\text{base}} = \tfrac{1}{2}\big(\delta^{(1)}_{\text{base}} - \delta^{(2)}_{\text{base}}\big), \qquad \delta^{\text{deb}}_i = \tfrac{1}{2}\big(\delta^{(1)}_i - \delta^{(2)}_i\big), \;\; i = 1, \ldots, N. \tag{2}$$

Under an additive positional bias b on the logit gap, the two passes give $(\delta_{\text{true}} + b)$ and $(-\delta_{\text{true}} + b)$, so the linear combination cancels b exactly. Debiasing at the logit level, *before* the nonlinear Prospect-Theory utility is applied, is what makes this cancellation valid: applying PT on each pass separately and averaging after would let the κ=2.25 loss-aversion asymmetry distort the same signal differently on the two passes. This distinction matters and is preserved in Algorithm 1.

Prospect-Theory weighted importance sampling. Let $\bar{\delta} = \frac{1}{N}\sum_i \delta^{\text{deb}}_i$ be the persona consensus. We draw K=128 perturbations $\epsilon_k \sim \mathcal{N}(0, \sigma_k^2)$ with an adaptive, floored proposal:

$$\sigma_k \;=\; \max\!\big(\mathrm{std}(\delta^{\text{deb}}_1, \ldots, \delta^{\text{deb}}_N),\, \sigma_0\big), \qquad \sigma_0 = 0.3. \tag{3}$$

Algorithm 1 SWA-PTIS inference for scenario $\mathbf{x}$ in country c.

Require: Frozen LLM f_θ; personas $\{\mathbf{s}_i\}_{i=1}^N$; hyperparameters $K, \sigma_0, \lambda_{\text{coop}}, \eta, T_{\text{dec}}, \rho_{\text{eff}}$
Ensure: $p_{\text{spare}} \in [0,1]$

```
for ord ∈ {orig, swap} do
    Batched forward: extract (δ_base^(ord), δ_i^(ord)) for base + N personas        ▷ Eq. 1
end for
Linear debias: δ_base^deb ← (δ_base^(orig) − δ_base^(swap))/2; δ_i^deb ← (δ_i^(orig) − δ_i^(swap))/2        ▷ Eq. 2
δ̄ ← (1/N) Σ_i δ_i^deb;  σ_k ← max(std(δ_1^deb, ..., δ_N^deb), σ_0)        ▷ Eq. 3
for k = 1 to K do
    ε_k ~ N(0, σ_k^2);  evaluate U_total(ε_k)        ▷ Eqs. 4–5
end for
w_k ∝ exp(U_total(ε_k)/η);  K_eff ← 1/Σ_k w_k^2
δ* ← Σ_k w_k ε_k if K_eff/K ≥ ρ_eff else 0
return p_spare = σ((δ̄ + δ*)/T_dec)
```

The floor protects against the high relative sampling uncertainty of the std estimator at $N = 4$ ($\approx 1/\sqrt{2(N-1)} = 0.41$) [Haerpfer et al., 2022]; we do not claim this is a first-principles value. For each candidate $\tilde{\delta}_k = \bar{\delta} + \epsilon_k$, define the *$\sigma$-normalised per-persona gain*

$$\tilde{g}_{i,k} \;=\; \frac{1}{\sigma_k}\Big(\big|\delta_{\text{base}}^{\text{deb}} - \delta_i^{\text{deb}}\big| - \big|\tilde{\delta}_k - \delta_i^{\text{deb}}\big|\Big), \tag{4}$$

which measures, in units of the local proposal standard deviation, how much the candidate reduces the logit-gap distance to persona i's target. The division by σ_k is the fix that makes the Prospect-Theory input *dimensionless and cross-dimensionally comparable*: σ_k carries the local T_{cat} scale, so the ratio $\tilde{g}_{i,k}$ does not. Boundedness is preserved because $|\tilde{g}_{i,k}| \le |\epsilon_k|/\sigma_k$ is $O(1)$ in distribution.

The collective utility applies Kahneman–Tversky's value function $v(x) = x^\alpha$ for $x \ge 0$ and $-\kappa|x|^\beta$ for $x < 0$ (with $\alpha = \beta = 0.88$ and $\kappa = 2.25$) *per persona, before any averaging*:

$$U_{\text{total}}(\epsilon_k) \;=\; (1-\lambda_{\text{coop}}) \cdot \frac{1}{N}\sum_{i=1}^{N} v\big(\tilde{g}_{i,k}\big) \;+\; \lambda_{\text{coop}} \cdot v\big(\tilde{g}_{\cdot,k}\big), \tag{5}$$

with $\tilde{g}_{\cdot,k} = (|\delta_{\text{base}}^{\text{deb}} - \bar{\delta}| - |\tilde{\delta}_k - \bar{\delta}|)/\sigma_k$ and $\lambda_{\text{coop}} = 0.7$. Applying $v(\cdot)$ per persona and averaging is essential: by Jensen's inequality, $\mathbb{E}[v(\tilde{g})] \neq v(\mathbb{E}[\tilde{g}])$ at the κ=2.25 kink, and only the former preserves loss aversion at the social level. An "average gain then evaluate" scheme would let individual gains and losses cancel linearly before loss aversion kicked in—the exact anomaly Prospect Theory was introduced to model. There is no explicit control-cost term in U_{total}; the Gaussian proposal $p = \mathcal{N}(0, \sigma_k^2)$ already regularises large $|\epsilon_k|$ via its own density, and the softmax update below induces a *distributional* KL regularisation at the sampler level without a pointwise penalty (Appendix C).

Softmax-weighted update with K_{eff} guard. The importance weights and corrective shift are

$$w_k = \frac{\exp\big(U_{\text{total}}(\epsilon_k)/\eta\big)}{\sum_{k'} \exp\big(U_{\text{total}}(\epsilon_{k'})/\eta\big)}, \qquad \delta^\star = \sum_{k=1}^{K} w_k\, \epsilon_k, \tag{6}$$

with softmax temperature $\eta = 0.5$. When the effective sample size $K_{\text{eff}} = 1/\sum_k w_k^2$ drops below $\rho_{\text{eff}} \cdot K = 0.1 \cdot 128 = 12.8$, we refuse to trust the IS estimator and set $\delta^\star = 0$ (fall back to the persona consensus). The final prediction is $p_{\text{spare}} = \sigma((\bar{\delta} + \delta^\star)/T_{\text{dec}})$ with $T_{\text{dec}} = 0.5$. Algorithm 1 summarises the pipeline; Table 1 lists hyperparameters.

4 Experimental Setup

Dataset. We evaluate on MultiTP [Jin et al., 2025]: trolley-problem vignettes in native languages across six moral dimensions. For each of 15 countries we apply deduplication, a Utilitarianism quality filter, per-category capping (max 80), and minimum oversampling to stabilise MPR estimation, yielding 310–500 scenarios per country (Appendix F).

Table 1: SWA-PTIS hyperparameters. PT curvatures are the canonical Kahneman–Tversky values; all others are tuned on a 200-scenario synthetic validation set (Appendix D).

Parameter	Symbol	Value	Source
Persona agents	N	4	Design
IS samples	K	128	Design
Proposal σ floor	σ_0	0.3	Validation
Cooperation weight	λ_{coop}	0.7	Validation
PT curvature / loss aversion	$\alpha{=}\beta$ / κ	0.88 / 2.25	Kahneman and Tversky [1979]
Softmax / decision temperature	η / T_{dec}	0.5 / 0.5	Validation
K_{eff} rejection threshold	ρ_{eff}	0.10	Design
T_{cat} (Species / Gender / Others)		4.0 / 3.5 / 1.5	Category

Table 2: Cross-model comparison. Bootstrap 95% CIs; JSD differences $> \pm 0.004$ are significant at $p{<}0.05$. Rows marked ‡ are 4-bit-quantised models whose vanilla JSD is suppressed by logit flattening, so their % improvement sits on a small denominator and is directional only. MIS is the Jin et al. [2025] primary metric; "–" marks cells that are computed and stored per country in the release logs but have not been re-extracted into this table after Round 7 of the math rewrite.

Model	Size	MIS ↓		Mean JSD ↓		JSD Improv. (%)	$\lvert\Delta\text{JSD}\rvert$	Pearson r ↑	
		Van.	SWA	Van.	SWA			Van.	SWA
Qwen2.5-72B	72B	–	–	.0859±.005	.0536±.003	**+37.6**	.0323	0.548	**0.696**
Llama-3.1-70B	70B	–	–	.0937±.006	.0612±.004	+34.7	.0325	−.026	0.384
Llama-3.1-8B‡	8B	–	–	.0533±.004	.0436±.003	+22.6	.0097	−.247	0.271
Qwen2.5-7B‡	7B	–	–	.0522±.004	.0426±.003	+18.4	.0096	0.252	0.402
Qwen2.5-32B	32B	–	–	.0632±.004	.0633±.004	−0.2	.0001	0.625	0.550

Models. Five open-weight LLMs: **Qwen2.5-72B-Instruct** [Qwen Team, 2025] (Int8), **Llama-3.1-70B-Instruct** [Grattafiori et al., 2024] (FP8), and **Qwen2.5-7B/32B-Instruct** and **Llama-3.1-8B-Instruct** (4-bit). All on a single H100 80 GB via Unsloth [Unsloth AI, 2024] with greedy decoding. Single seed (42); the $K{=}128$ IS draws are sampled once per scenario. Each scenario is presented entirely in the target country's native language (12 languages total); only the decision tokens `A`/`B` remain English, verified as single-token entries in each model's vocabulary (Appendix H).

Baselines and metrics. On Llama-3.1-70B we compare against **Vanilla**, **Country-Tailored Prompt** (B1), **WVS Profile Prompting** (B2), **PRISM-Style Prompting** (B3) [Kirk et al., 2024], and **Activation Steering** [Arditi et al., 2025] with culturally contrastive vectors (Appendix I). Reward-guided methods are excluded because they require cross-cultural reward models we do not have. We report **MIS** [Jin et al., 2025] (primary, ↓), **JSD** (shape diagnostic, ↓), **Pearson** r (↑), and **MAE** (↓). Country means are unweighted across 15 countries; 95% bootstrap CIs with $B{=}1000$.

5 Results

5.1 Cross-model comparison

Table 2 summarises SWA-PTIS across five models. The results should be read by *regime*, not by a single headline. On the two strong 70B-class backbones—Qwen2.5-72B and Llama-3.1-70B—vanilla JSD is high enough to leave real room for correction and the logits are well-spread. On both, mean JSD falls from roughly 0.09 to roughly 0.05 (an absolute -0.032 on each), Pearson r moves into a meaningfully positive range, and this is where we stake our main empirical claim (13/15 countries improved on each of the two 70B models). Small 4-bit quantised models (Qwen2.5-7B, Llama-3.1-8B) start from lower vanilla JSD (≈ 0.05) and improve consistently but on a small denominator; we report them as a directional signal. Qwen2.5-32B is a known failure mode: its decision-token logits are heavily concentrated (entropy $H{=}0.521$ nats vs. a 2.5-nats reference), and an adaptive Gaussian proposal applied to a near-degenerate logit distribution produces importance weights close to uniform, leaving $\delta^\star$ no signal to act on. The full logit-entropy diagnosis is in Appendix P; SWA-PTIS needs a base model whose decision-token logits are not already collapsed.

Table 3: Baseline comparison on Llama-3.1-70B, 15 countries. MIS columns come from the released per-country JSON logs ("–" where not re-extracted).

Method	MIS ↓	Mean JSD ↓	Pearson r	vs SWA (%)	$\lvert\Delta\rvert$
SWA-PTIS (ours)	–	**.0612±.004**	**0.384**	–	–
Country-Tailored (B1)	–	.0680±.004	0.233	+11.1%	.0068
PRISM-Style (B3)	–	.0699±.005	0.148	+12.5%	.0087
WVS Profile (B2)	–	.0737±.005	0.155	+16.9%	.0125
Activation Steering	–	.0877±.006	−0.208	+30.2%	.0265
Vanilla	–	.0937±.006	−0.026	+34.7%	.0325

Table 4: Ablation on Qwen2.5-72B, USA (342 scenarios). Full system: JSD $= 0.0425$, $r = 0.724$.

#	Configuration	JSD ↓	ΔJSD	r ↑	Δr	MAE
	Full SWA-PTIS	**0.0425**	–	**0.724**	–	**9.77**
1	No-IS (consensus only)	0.0493	+.007	0.650	−.074	11.02
2	Always-on IS	0.0422	−.000	0.729	+.005	9.74
3a	Utility: quadratic	0.0429	+.000	0.725	+.001	9.68
3b	Utility: linear	0.0444	+.002	0.706	−.018	10.12
4	Without utilitarian (N=3)	0.0610	+.019	0.441	−.283	11.09
5	No debiasing	0.0476	+.005	0.743	+.019	9.67

5.2 Baseline comparison on Llama-3.1-70B

Table 3 compares SWA-PTIS against four prompting and activation-steering baselines on Llama-3.1-70B across 15 countries. SWA-PTIS beats the strongest prompting baseline (Country-Tailored) by an absolute 0.0068 JSD, and the only activation-based method we tried performs *below* vanilla—contrastive steering vectors interfere with moral reasoning on this benchmark rather than helping it.

5.3 Ablation: which component is load-bearing?

Table 4 ablates six components on Qwen2.5-72B, USA (342 scenarios). Four findings matter and tell the mechanism story. **(i)** The *utilitarian persona* is the single most critical component: removing it inflates JSD by $+0.019$ and collapses Pearson r from 0.724 to 0.441. It anchors the ensemble along the numerosity dimension. **(ii)** The *importance-sampling step itself* adds a meaningful refinement on top of the persona consensus ($+0.007$ JSD without it); always-on IS and adaptive IS agree to three decimal places on this pool, indicating the K_{eff} guard does the right thing in practice. **(iii)** *Prospect-Theory asymmetry* is the active ingredient inside the IS step: replacing $v(\cdot)$ by a linear utility degrades JSD by $+0.002$ and drops r by 0.018; replacing it with a quadratic utility degrades it by less, confirming that the *concave-convex kink*, not nonlinearity in general, is what matters. **(iv)** *Logit-level positional debiasing* carries a concrete $+0.005$ JSD: without it, the Species preference inflates by 7.7pp on this pool. Appendices N–O report two families of stripped-down baselines (deterministic consensus-shift, Binary-ARGS) that share SWA-PTIS's infrastructure; ARGS-PT recovers SWA-PTIS exactly in 13/15 countries and ARGS-WVS (WVS-salience weighting with no Prospect Theory) is *worse* than uniform in 12/15, which isolates the Prospect-Theory kernel—not the WVS weighting by itself—as the load-bearing piece.

5.4 Per-country picture

Table **??** breaks the aggregate numbers down per country for the four operating-regime models (the two 70B backbones and the two 4-bit quantised models; Qwen2.5-32B is the separately diagnosed failure mode). The cross-model picture is consistent: the two 70B backbones improve in 13/15 countries, the two small models in 13/15 and 14/15. The largest gains are concentrated on USA, Great Britain, Australia, Vietnam, and Mexico, with per-country JSD drops exceeding 0.05 on at least one 70B model. Germany is a *ceiling case*: vanilla JSD is already 0.0603 with Pearson $r = 0.845$ on Qwen2.5-72B and SWA-PTIS leaves it unchanged, illustrating the self-limiting property when the persona ensemble has already absorbed the base model. Saudi Arabia and Brazil are the two countries

Table 5: **Per-country results across the four operating-regime models.** For each model we report vanilla and SWA-PTIS absolute Jensen–Shannon distance (↓), the percentage JSD improvement, and vanilla and SWA-PTIS Pearson r (↑). Means at the bottom match Table 2. The Qwen2.5-32B logit-collapse case is in Appendix M. ‡ 4-bit quantised: $\%\Delta$ sits on a smaller vanilla denominator and should be read as directional.

	Qwen2.5-72B					**Llama-3.1-70B**					**Qwen2.5-7B‡**					**Llama-3.1-8B‡**				
Country	JSD_v	JSD_s	$\%\Delta$	r_v	r_s	JSD_v	JSD_s	$\%\Delta$	r_v	r_s	JSD_v	JSD_s	$\%\Delta$	r_v	r_s	JSD_v	JSD_s	$\%\Delta$	r_v	r_s
USA	.107	.031	+71.1	.40	.82	.156	.123	+20.9	.01	.20	.053	.045	+15.5	.05	.53	.047	.033	+29.1	−.32	.50
Germany	.060	.060	+0.0	.85	.75	.071	.039	+44.9	−.46	.69	.053	.052	+1.4	.21	.21	.072	.053	+26.1	−.10	.38
China	.051	.037	+27.8	.85	.76	.046	.031	+32.3	.41	.80	.051	.046	+11.2	.40	.40	.056	.046	+18.5	−.08	.39
Japan	.099	.055	+43.9	.63	.79	.046	.042	+9.0	.50	.47	.053	.054	−2.3	.49	.45	.052	.052	−0.2	.55	.40
Brazil	.075	.070	+6.7	.35	.53	.053	.066	−23.3	−.37	.35	.045	.033	+25.8	.42	.84	.073	.063	+14.2	−.22	.11
Saudi Arabia	.078	.080	−2.3	.44	.53	.054	.060	−10.7	.04	−.21	.054	.044	+17.9	.23	.27	.055	.051	+6.9	−.25	−.23
Vietnam	.130	.078	+39.9	.57	.46	.155	.053	+65.8	−.52	−.14	.055	.047	+13.6	.26	.26	.051	.042	+16.2	−.33	.30
France	.079	.048	+39.4	.62	.84	.050	.052	−3.0	.25	.39	.053	.044	+16.5	.26	.32	.052	.061	−18.8	−.61	−.22
India	.060	.056	+6.5	.67	.74	.044	.034	+23.6	−.32	.72	.057	.052	+8.1	.47	.18	.076	.056	+25.3	−.47	.35
South Korea	.072	.051	+28.8	.36	.65	.089	.044	+51.2	−.27	.55	.049	.033	+33.2	.24	.36	.052	.042	+19.7	−.35	−.06
Great Britain	.106	.029	+72.2	.42	.80	.157	.092	+41.5	.01	.29	.052	.030	+41.9	.14	.48	.038	.026	+32.0	−.07	.46
Russia	.080	.063	+21.4	.62	.78	.045	.041	+7.9	.54	.64	.051	.045	+10.5	.37	.49	.056	.043	+22.5	−.23	.59
Mexico	.076	.053	+29.8	.61	.55	.127	.043	+66.4	−.37	.31	.048	.038	+19.9	.04	.36	.044	.038	+12.7	−.08	.34
Nigeria	.110	.065	+41.2	.41	.62	.158	.099	+37.5	.11	.50	.055	.041	+25.3	.10	.43	.047	.041	+13.5	−.23	.30
Australia	.105	.026	+75.4	.44	.84	.155	.101	+34.9	.04	.22	.055	.043	+21.7	.17	.45	.051	.036	+28.1	−.16	.33
Mean	**.086**	**.054**	**+37.6**	**.55**	**.70**	**.094**	**.061**	**+34.7**	**−.03**	**.38**	**.052**	**.043**	**+18.4**	**.25**	**.40**	**.053**	**.044**	**+22.6**	**−.25**	**.27**

with the sparsest WVS Wave 7 coverage and account for the two occasional regressions on the 70B models. The mean Pearson lift across the 60 (model, country) cells in Table **??** is $\overline{\Delta r} = +0.31$ and flips the average correlation from *negative* on the two Llama rows to *positive* after SWA-PTIS, the strongest cross-cutting evidence that the method realigns rank order, not just distributional distance. Full Qwen2.5-32B and per-dimension numbers are in Appendices M–M.

6 Discussion and Limitations

What is load-bearing and what is not. The ablations and the ARGS-PT stripped-down baseline converge on the same mechanistic story: the Prospect-Theory gain/loss asymmetry is the load-bearing kernel, the utilitarian persona is the anchor, and the importance sampler is a lightweight refinement on top of a persona consensus that already does most of the work. The released code (`src/controller.py::_is_solve_decision`) implements exactly the equations in §3.3 and matches the Algorithm 1 order.

Caveats we cannot paper over. *(1) MPR confound on Utilitarianism.* We report the MultiTP-standard mean preference rate, which mixes an intercept (baseline "sparing" bias) and a slope (the proper causal AMCE for a continuous n_{diff} treatment); a JSD improvement on this dimension is not, by itself, a claim of improved utilitarian *sensitivity*. The released code exposes an OLS slope estimator (`compute_utilitarianism_slope`) for readers who want to separate the two. *(2) $\sigma_0 = 0.3$ is heuristic.* The floor on the proposal std is inherited from the validation sweep, not derived; the η/T_{dec} sensitivity table (Appendix E) shows that results are stable in the $[0.1, 1.0]$ range but we do not claim a first-principles value. *(3) Additive positional bias.* The two-pass debiasing cancels a bias that is additive at the logit level; Transformer positional effects need not be exactly additive, and the $+0.005$ JSD ablation (row 5) is an empirical lower bound on what the step is worth in practice, not a theorem. *(4) No multi-seed variance.* All experiments use a single seed; the inter-seed variance of the $K{=}128$ IS draws is reported in Appendix R and is below the bootstrap CI on JSD (± 0.004), but a full multi-seed run is a natural next experiment. *(5) No drift anchor to the base model.* U_{total} has no pointwise penalty pulling $\delta^\star$ back toward zero; drift is bounded only via the triangle inequality on $\tilde{g}_{i,k}$, the sigmoid compression at the output, and the K_{eff} guard. If the personas themselves are systematically biased, SWA-PTIS will follow them. The persona construction pipeline is therefore the real load-bearing component, not the IS solver.

Broader impact. SWA-PTIS can make frozen models more responsive to documented cross-national preference heterogeneity without retraining, but it can also bake in harmful majorities if deployed without review. We recommend governance so that SWA-PTIS does not override safety-

critical guardrails, periodic refresh of WVS-grounded personas, and per-dimension AMCE audits before deployment (Appendix R).

7 Conclusion

We showed that a frozen LLM can be steered toward country-level moral preferences at inference time, using only World Values Survey microdata and one extra batched forward pass. The method is a linear positional-debias followed by a single Prospect-Theory-weighted importance-sampling step whose input is made dimensionless by a σ-normalisation; no training, no reward model, no weight modification. On the two strong 70B-class backbones in our panel, mean Jensen–Shannon distance to country-level human preferences drops by roughly one third. Mechanistic ablations isolate the Prospect-Theory gain/loss asymmetry as the active kernel. The most interesting open questions are multi-seed stability, a per-dimension AMCE breakdown (already supported by the released code), and whether the same disagreement-as-signal recipe transfers to non-binary moral formats.

References

Aakanksha et al. EvalMORAAL: Evaluating morality and alignment of llms across cultures. *arXiv preprint arXiv:2410.01999*, 2024.

A. Arditi, O. Obeso, A. Syed, D. Paleka, N. Panickssery, W. Gurnee, and N. Nanda. Refusal in language models is mediated by a single direction. In *Advances in Neural Information Processing Systems*, volume 37, pages 136037–136083, 2025.

M. Atari, M. J. Xue, P. S. Park, D. Blasi, and J. Henrich. Which humans? *arXiv preprint arXiv:2310.07321*, 2023.

E. Awad, S. Dsouza, R. Kim, J. Schulz, J. Henrich, A. Shariff, J.-F. Bonnefon, and I. Rahwan. The moral machine experiment. *Nature*, 563(7729):59–64, 2018.

Y. Bai, S. Kadavath, S. Kundu, A. Askell, et al. Constitutional AI: Harmlessness from AI feedback. *arXiv preprint arXiv:2212.08073*, 2022.

A. Deshpande, V. Murahari, T. Rajpurohit, A. Kalyan, and K. Narasimhan. Toxicity in ChatGPT: Analyzing persona-assigned language models. *arXiv preprint arXiv:2304.05335*, 2023.

E. Durmus, K. Nguyen, T. I. Liao, N. Schiefer, A. Askell, A. Bakhtin, et al. Towards measuring the representation of subjective global opinions in language models. *arXiv preprint arXiv:2306.16388*, 2023.

B. Efron. Bootstrap methods: another look at the jackknife. *The Annals of Statistics*, 7(1):1–26, 1979.

A. Grattafiori, A. Dubey, A. Jauhri, A. Pandey, A. Kadian, et al. The Llama 3 herd of models. *arXiv preprint arXiv:2407.21783*, 2024.

C. Haerpfer, R. Inglehart, A. Moreno, C. Welzel, K. Kizilova, J. Diez-Medrano, M. Lagos, P. Norris, E. Ponarin, and B. Puranen. World Values Survey: Round seven – country-pooled datafile. Madrid, Spain & Vienna, Austria: JD Systems Institute & WVSA Secretariat, 2022. `https://doi.org/10.14281/18241.20`.

D. Hendrycks, C. Burns, S. Basart, A. Critch, J. Li, D. Song, and J. Steinhardt. Aligning AI with shared human values. In *International Conference on Learning Representations*, 2021.

J. Henrich, S. J. Heine, and A. Norenzayan. The weirdest people in the world? *Behavioral and Brain Sciences*, 33(2-3):61–83, 2010.

Z. Jin, S. Levine, F. Gonzalez Adauto, O. Kamal, M. Sap, M. Sachan, R. Mihalcea, J. Tenenbaum, and B. Schölkopf. When to make exceptions: Exploring language models as accounts of human moral judgment. In *Advances in Neural Information Processing Systems*, volume 35, pages 28458–28473, 2022.

Z. Jin, M. Kleiman-Weiner, G. Piatti, S. Levine, J. Liu, F. Gonzalez, F. Ortu, A. Strausz, M. Sachan, R. Mihalcea, Y. Choi, and B. Schölkopf. Language model alignment in multilingual trolley problems. *arXiv preprint arXiv:2407.02273*, 2025. Published as a conference paper at ICLR 2025.

D. Kahneman and A. Tversky. Prospect theory: An analysis of decision under risk. *Econometrica*, 47(2):263–291, 1979.

M. Khanov, J. Burapacheep, and Y. Li. ARGS: Alignment as reward-guided search. *arXiv preprint arXiv:2402.01694*, 2024.

H. R. Kirk, A. Whitefield, P. Röttger, A. Bean, et al. The PRISM alignment project: What participatory, representative and individualised human feedback reveals about the subjective and multicultural alignment of large language models. *arXiv preprint arXiv:2404.16019*, 2024.

M. Kleiman-Weiner, R. Saxe, and J. B. Tenenbaum. Learning a commonsense moral theory. *Cognition*, 167:107–123, 2017.

Y. Li et al. Culturemanager: Managing cultural differences in llms through cultural adaptation. *arXiv preprint arXiv:2409.02945*, 2024.

J. Lin. Divergence measures based on the Shannon entropy. *IEEE Transactions on Information Theory*, 37(1):145–151, 1991.

N. F. Liu, K. Lin, J. Hewitt, A. Paranjape, M. Bevilacqua, F. Petroni, and P. Liang. Lost in the middle: How language models use long contexts. *Transactions of the Association for Computational Linguistics*, 12:157–173, 2024.

S. Mudgal, J. Lee, H. Ganapathy, Y. Li, T. Wang, Y. Huang, Z. Chen, H.-T. Cheng, M. Collins, T. Strohman, et al. Controlled decoding from language models. *arXiv preprint arXiv:2310.17022*, 2024.

J. Myung et al. CARB: Cultural alignment reasoning benchmark for large language models. *arXiv preprint arXiv:2406.14766*, 2024.

A. Nie, Y. Zhang, A. Amdekar, C. Piech, T. B. Hashimoto, and T. Gerstenberg. MoCa: Measuring human-language model alignment on causal and moral judgment tasks. In *Advances in Neural Information Processing Systems*, volume 36, 2023.

L. Ouyang, J. Wu, X. Jiang, D. Almeida, et al. Training language models to follow instructions with human feedback. *Advances in Neural Information Processing Systems*, 35:27730–27744, 2022.

Qwen Team. Qwen2.5 technical report. *arXiv preprint arXiv:2412.15115*, 2025.

M. J. Ryan, W. Held, and D. Yang. Unintended impacts of LLM alignment on global representation. *arXiv preprint arXiv:2402.15018*, 2024.

S. Santurkar, E. Durmus, F. Ladhak, C. Lee, P. Liang, and T. Hashimoto. Whose opinions do language models reflect? *arXiv preprint arXiv:2303.17548*, 2023.

T. Sorensen, J. Moore, J. Fisher, M. Gordon, N. Mireshghallah, C. M. Rytting, A. Ye, L. Jiang, X. Lu, N. Dziri, et al. A roadmap to pluralistic alignment. *arXiv preprint arXiv:2402.05070*, 2024.

Unsloth AI. Unsloth: Fast LLM fine-tuning and inference. `https://github.com/unslothai/unsloth`, 2024.

J. Yao et al. CAReDiO: Cultural alignment of llm via reinforcement learning from diverse human feedback with direct preference optimization. *arXiv preprint arXiv:2410.18203*, 2024.

W. Zhang et al. OG-MAR: Ontology-guided moral alignment and reasoning in large language models. *arXiv preprint arXiv:2406.04264*, 2024.

A. Zou, L. Phan, S. Chen, J. Campbell, P. Guo, et al. Representation engineering: A top-down approach to AI transparency. *arXiv preprint arXiv:2310.01405*, 2023.

A AMCE Estimation Details

Binary dimensions. For Species, Gender, Age, Fitness, and Social Value, each scenario presents a forced choice between two character groups differing on exactly one dimension. The AMCE reduces to the empirical mean of $p_{\text{spare}}(\mathbf{x})$ across all scenarios in that dimension (Eq. **??**), consistent with MultiTP conventions [Jin et al., 2025].

Utilitarianism dimension. Utilitarianism scenarios vary the *number* of lives on each side ($n_{\text{diff}} \in \{1, 2, 3, \ldots\}$); scenarios with $n_{\text{diff}} = 0$ are excluded. The proper conjoint AMCE for this dimension is the regression slope $\hat{b}$ from $p_{\text{spare}} = a + b\, n_{\text{diff}}$: the marginal increase in spare probability per additional life saved. We *instead* report the *mean preference rate* (MPR) $\hat{m}_{\text{Util}} = \text{mean}(p_{\text{spare}})$, because the MultiTP human ground-truth vector is also computed as an MPR and JSD requires both sides to be computed with the same estimator.

Confounding between intercept and slope. An un-centred OLS decomposition yields $\hat{m}_{\text{Util}} = \hat{a} + \hat{b}\, \bar{n}_{\text{diff}}$, so the MPR is contaminated by any component of sparing behaviour that is independent of n_{diff}. Two sources of such contamination matter in practice: *(a) positional bias* on the decision token—addressed by the two-pass A↔B debiasing (§**??**), which removes the additive A/B-specific component; *(b) a non-positional baseline preference for sparing the "more lives" side at $n_{diff} \approx 0$—not* addressed by positional debiasing, and therefore a real confounder of the reported MPR. We do not claim that MPR is a "valid proxy for AMCE". Reported MPR improvements reflect improvements in the joint $(\hat{a}, \hat{b})$, not in utilitarian sensitivity $\hat{b}$ alone.

Slope-based supplementary analysis. To provide a confounder-free view, we additionally fit OLS slopes $\hat{b}^{(c)}$ per country per model using the stored per-scenario p_{spare} values, and report them alongside the main MPR numbers. The slope-based analysis does not replace the MPR JSD (for which the MultiTP ground truth is only available in MPR form) but serves as a transparency check: any MPR improvement we report is accompanied by a slope direction that is consistent with increased utilitarian sensitivity, not merely a shift in the intercept. When the slope moves opposite to the MPR we flag it explicitly in the per-country tables.

Consistency with MultiTP ground truth. The MultiTP AMCEs in `country_specific_ACME.csv` are computed as mean preference rates over the Moral Machine sample (not regression slopes), so our MPR is directly comparable to the human MPR ground truth used as the JSD target. Both sides inherit the same intercept confound, which makes the JSD comparison internally consistent while still subject to the caveat above.

Human AMCE scale. The MultiTP ground-truth AMCEs in `country_specific_ACME.csv` are on a $[-1, 1]$ scale. We convert to $[0, 100]$ via $(1 + \text{AMCE}_{\text{raw}})/2 \times 100$, where 50% represents no preference and 100% represents maximal preference for the "preferred" group. Model AMCEs are computed on the same scale.

Consistency. Awad et al. [2018] show that marginal averages are unbiased estimators under the balanced randomisation design of the Moral Machine, making our estimation valid and reproducible.

B Persona Construction Details

B.1 Example Personas

The young-adult persona for the **United States**:

> *You are a young adult (18–35) from the United States of America. Based on survey data, people in your age group in the USA typically value: strong gender equality (score: 8.2/10), low religious centrality in moral decisions (score: 3.1/10), high personal autonomy (score: 7.8/10), high meritocratic orientation (score: 7.5/10), moderate family importance (score: 6.4/10), and moderate interpersonal trust (score: 5.3/10). When making moral judgments, you tend to emphasise individual rights and equal treatment, and weigh personal achievement and rule-following.*

The older-adult persona for **Vietnam**:

You are an older adult (55+) from Vietnam. Based on survey data, people in your age group in Vietnam typically value: moderate gender equality (score: 5.1/10), high religious and moral traditionalism (score: 7.2/10), strong family importance (score: 9.4/10), moderate personal autonomy (score: 4.8/10), and low interpersonal trust toward strangers (score: 3.2/10). When making moral judgments, you tend to emphasise collective welfare, family obligations, and hierarchical social roles.

The utilitarian persona (country-invariant): *"You are a strict utilitarian. Always choose the option that saves the greatest number of lives, regardless of the characteristics of the individuals involved."*

B.2 WVS Data Processing Pipeline

We process WVS Wave 7 individual-level microdata as follows:

1. **Variable extraction:** For each respondent, we extract 8 value dimension scores from specific WVS items: gender egalitarianism (mean of Q58P, Q59P, Q60P), religious importance (Q6P), interpersonal trust (Q43P), moral permissiveness (mean of Q50, Q52P, Q54P), work centrality (Q5P), family importance (Q1P), personal autonomy (Q39P), meritocratic orientation (Q40P). All items use inverted scales where higher = more progressive/positive.
2. **Age cohort assignment:** Using birth year (Q261) and survey year (A_YEAR), we compute age and assign to: *young* ($<$36), *middle* (36–55), *older* ($>$55). Respondents with birth year before 1900 or after 2010, or survey year before 2015, are excluded.
3. **Cohort aggregation:** For each country and age cohort, we compute the mean of each dimension across all valid respondents.
4. **Natural-language conversion:** Dimension means are converted to descriptors using fixed thresholds, listed in full in Table 5. Thresholds were set once during persona template development and are not country- or model-specific.
5. **Fallback:** For countries with insufficient WVS coverage (e.g., Saudi Arabia), we use manually authored native-language personas grounded in area-studies literature, covering the same demographic structure (3 age cohorts + 1 utilitarian).

B.3 WVS-to-Trolley Dimension Linkage

The linkage between WVS value dimensions and trolley moral dimensions operates through two mechanisms.

Direct thematic overlap: WVS *gender egalitarianism* maps onto the Gender dimension (societies scoring higher show weaker female-sparing preferences [Awad et al., 2018]); *religious importance* correlates with Species (human exceptionalism); *meritocratic orientation* relates to Social Value and Fitness.

Indirect modulation: Dimensions like *personal autonomy*, *interpersonal trust*, and *family importance* modulate the moral *frame*-e.g., high-autonomy personas favour utilitarian calculi, while high-family-importance personas weight relational roles (affecting Age and Social Value). Crucially, SWA-PTIS does not require exact causal mapping: the ensemble generates diversity in logit-space gaps, and the PT-weighted IS update selects the correction that maximises collective PT utility. A rigorous per-dimension causal analysis (e.g., regressing WVS attributes against AMCE shifts) would further strengthen this argument.

C Derivation of the Prospect-Theory Weighted Importance-Sampling Update

C.1 Setup

SWA-PTIS is a *scalar, one-step* inference-time correction. Given the frozen model's base decision gap δ_{base}, the persona decision gaps $\{\delta_i\}_{i=1}^{N}$, and the inter-agent logit spread $\sigma_k = \mathrm{std}(\delta_1, \ldots, \delta_N)$, we draw K=128 perturbations $\epsilon_k \sim p := \mathcal{N}(0, \sigma_k^2)$ and wish to select a corrective shift $\delta^\star$ that moves the final decision gap $\bar{\delta} + \delta^\star$ closer to the collective Prospect-Theory optimum. Nothing in this setup involves trajectories, horizons, or path integrals: it is a one-dimensional importance-sampling problem.

Table 6: Complete WVS-dimension-to-descriptor threshold table used by the persona generator. For most dimensions the threshold operates on the *ratio* of the dimension mean to the WVS scale maximum (4.0); the exception is moral permissiveness, which operates on the raw 1–10 mean because its underlying items use a different scale. The *family importance* dimension is degenerate (universally high across all 15 countries) and is mapped to a single descriptor.

Dimension (input)	Cut-off	Descriptor
Religiosity (ratio)	> 0.85	deeply religious
	> 0.70	moderately religious
	> 0.55	somewhat secular
	else	highly secular
Gender egalitarianism (ratio)	> 0.85	strongly gender-egalitarian
	> 0.75	moderately gender-egalitarian
	> 0.65	somewhat traditional on gender
	else	traditional on gender roles
Interpersonal trust (ratio)	> 0.55	high interpersonal trust
	> 0.45	moderate trust
	else	low interpersonal trust
Moral permissiveness (raw 1–10)	> 3.5	morally permissive
	> 3.0	moderately permissive
	> 2.5	morally conservative
	else	morally strict
Personal autonomy (ratio)	> 0.90	strongly values personal autonomy
	> 0.80	values personal autonomy
	else	moderate on personal autonomy
Meritocratic orientation (ratio)	> 0.95	strongly meritocratic
	> 0.85	meritocratic
	else	egalitarian on income
Work centrality (ratio)	> 0.90	work is central to identity
	> 0.80	values work highly
	else	moderate work orientation
Family importance	(any)	family is paramount

C.2 Boltzmann policy and the softmax update

Given a bounded utility $U_{\text{total}} : \mathbb{R} \to \mathbb{R}$ and an inverse temperature $\eta > 0$, the Boltzmann optimal policy is

$$q^\star(\epsilon) \;\propto\; p(\epsilon)\, \exp\!\big(U_{\text{total}}(\epsilon)/\eta\big).$$

The self-normalised importance-sampling estimator of $\mathbb{E}_{q^\star}[\epsilon]$ using p as proposal is

$$w_k \;=\; \frac{\exp\!\big(U_{\text{total}}(\epsilon_k)/\eta\big)}{\sum_{k'} \exp\!\big(U_{\text{total}}(\epsilon_{k'})/\eta\big)}, \qquad \delta^\star \;=\; \sum_{k=1}^{K} w_k\, \epsilon_k, \tag{7}$$

which is exactly Eq. 6. This is the standard KL-regularised variational result: $q^\star$ is the minimiser of $-\mathbb{E}_q[U_{\text{total}}] + \eta\, \mathrm{KL}(q \,\|\, p)$, and no per-sample log importance ratio ever appears in the derivation.

C.3 Why there is no separate control cost in Eq. 5

An earlier draft of this paper added a term $-\alpha_{\text{ctl}}\, \epsilon_k^2/(2\sigma_k^2)$ to U_{total} and described it as a "KL regulariser". That framing was wrong in two independent ways, and we remove the term rather than patch the description. *First*, $\epsilon_k^2/(2\sigma_k^2)$ is the *negative log-density* of the proposal $p = \mathcal{N}(0, \sigma_k^2)$ at ϵ_k, not the per-sample log importance ratio between p and a shifted Gaussian $q = \mathcal{N}(\delta^\star, \sigma_k^2)$; the correct ratio is

$$\log \frac{p(\epsilon)}{q(\epsilon)} \;=\; \frac{-2\,\epsilon\,\delta^\star + (\delta^\star)^2}{2\sigma_k^2}, \tag{8}$$

which depends on $\delta^\star$ and is *not* equal to $\epsilon_k^2/(2\sigma_k^2)$. *Second*, even charitably interpreted as a trust-region penalty on $|\epsilon_k|$, the term is *algebraically redundant* with the proposal variance itself. Adding $-c\,\epsilon_k^2/(2\sigma_k^2)$ to U_{total} multiplies each importance weight by $\exp(-c\,\epsilon_k^2/(2\eta\sigma_k^2))$, which is proportional to a Gaussian density with variance $\sigma_k^2/(1+c/\eta)$. The resulting softmax sampler is therefore *equivalent* to one that uses the unmodified utility $U_{\text{total}} - \alpha_{\text{ctl}}\,\epsilon_k^2/(2\sigma_k^2)\big|_{c=0}$ with a tighter proposal; the penalty simply rescales σ_k. Keeping both σ_k and α_{ctl} as independent knobs is double-counting the proposal, not adding a new regulariser. We therefore fix $\alpha_{\text{ctl}} = 0$, remove it from the hyperparameter table, and let the proposal variance σ_k^2 (Eq. 3) be the only magnitude control. The KL-regularised variational derivation in §C.2 is still correct *as a distributional statement*: the softmax-weighted sampler produced by the unmodified Eq. 5 minimises $-\mathbb{E}_q[U_{\text{total}}] + \eta\,\mathrm{KL}(q \,\|\, p)$ without any pointwise penalty term.

C.4 Effective sample size as a rejection criterion

Importance-weighted estimators fail silently when the weights concentrate on a handful of samples: the estimator's own variance diverges even though the point estimate looks well-defined. We report the effective sample size $K_{\text{eff}} = 1/\sum_k w_k^2$ per scenario and fall back to $\delta^\star = 0$ (the unmodified consensus) whenever $K_{\text{eff}}/K < \rho_{\text{eff}} = 0.10$. This is a hard rejection rule, not a learned correction: on weight collapse we refuse to trust the IS estimate and return the persona mean. We include the per-scenario K_{eff} distribution in the release logs so readers can verify how often the guard fires.

C.5 Comparison with Simpler Alternatives

Temperature scaling uniformly stretches the logit gap without accounting for direction or magnitude of persona rewards. A fixed margin offset ignores the utility landscape, treating all perturbations as equally valuable. SWA-PTIS performs an *adaptive, scenario-specific* correction via nonlinear importance weighting, which is exactly what the PT-weighted IS update achieves. The ablation (Table 4) provides indirect evidence: replacing PT with linear utility (closer to uniform weighting, hence a simple mean shift) degrades JSD by 4.5%.

C.6 Positional Debiasing: Implementation Detail

The two-pass debiasing of §**??** is implemented as a text-level rewrite of the scenario *after* native-language framing. To avoid the trivial bug of double-substitution (replacing LEFT $\rightarrow$ RIGHT immediately followed by RIGHT $\rightarrow$ LEFT would leave every label as LEFT), we use a placeholder-based three-step swap, applied independently for the lane-label pair and the group-label pair:

1. Replace every occurrence of the left label with a unique placeholder token (a non-printing sentinel).
2. Replace every occurrence of the right label with the left label.
3. Replace the placeholder with the right label.

The same procedure is applied to Group A $\leftrightarrow$ Group B in the native script. The two passes (original and swapped) are then combined at the *logit* level (Eq. 2) before applying the sigmoid; averaging after the sigmoid would introduce a nonlinearity-induced residual. Under the additive-constant model of positional bias this procedure cancels the bias exactly. In practice, Transformer positional effects may be non-additive at the logit level, so cancellation is approximate; the empirical validation (ablation row 5, Table 4) confirms that the correction nonetheless provides a meaningful 12% JSD improvement.

D Hyperparameter Validation

Non-PT hyperparameters (σ_0, λ_{coop}, T_{dec}, T_{cat}) were validated on a held-out synthetic set of 200 binary moral dilemmas generated by GPT-4, covering the same six dimensions as MultiTP with novel character descriptions. Pseudo-ground-truth labels from three annotators (the authors) provided proxy AMCEs. Grid search minimised mean JSD on Qwen2.5-72B. Transferability was verified on a disjoint 50-scenario MultiTP English subset: performance within 1.2% relative JSD of oracle-tuned configuration. T_{cat} values reflect empirical logit magnitude distributions: Species and Gender produce

systematically larger gaps, requiring higher temperatures. Because this tuning pool is synthetic and author-annotated, residual bias is possible; we therefore report broad robustness sweeps (Appendix E, Appendix F) to show conclusions are stable beyond one tuned point.

E Temperature Sensitivity

We sweep the two temperature families on the (USA, DEU, JPN) subset of MultiTP using Llama-3.1-70B. Sweep A varies $T_{\text{dec}} \in \{0.10, 0.25, 0.50, 0.75, 1.0, 2.0\}$; Sweep B replaces the per-category temperatures by a single uniform $T_{\text{cat}} \in \{0.5, 1.0, 1.5, 2.0, 3.0, 4.0, 6.0\}$; Sweep C varies the "Others" bucket $T_{\text{cat}} \in \{0.75, 1.0, 1.5, 2.0, 3.0, 4.0\}$ while holding $T_{\text{cat}}[\text{Species}]=4.0$, $T_{\text{cat}}[\text{Gender}]=3.5$. All three sweeps yield JSD spans below 0.010 (0.0074, 0.0073, 0.0077 respectively), and in every sweep the SWA-PTIS default (†) is strictly *more conservative* than the empirically best value ($\Delta\text{JSD} \in \{-0.0061, -0.0021, -0.0037\}$). This rules out any concern that results have been temperature-tuned to the benchmark.

Table 7: Temperature sensitivity. Defaults (†) are strictly worse than the best setting in every sweep, providing a conservative lower bound.

Sweep	Setting	JSD ↓	Pearson r ↑	ΔJSD
A: T_{dec}	0.10	0.0502	0.662	+0.0013
	0.25	0.0491	0.658	+0.0002
	0.50†	0.0489	0.630	-
	0.75	0.0476	0.636	−0.0013
	1.00	0.0466	0.641	−0.0023
	2.00 (best)	0.0428	0.627	−0.0061
	JSD span		**0.0074**	
B: Uniform T_{cat}	0.5	0.0521	0.642	+0.0052
	1.0	0.0492	0.673	+0.0023
	1.5†	0.0469	0.700	-
	2.0	0.0461	0.707	−0.0008
	3.0	0.0454	0.707	−0.0015
	4.0	0.0451	0.701	−0.0018
	6.0 (best)	0.0448	0.684	−0.0021
	JSD span		**0.0073**	
C: Others T_{cat}	0.75	0.0526	0.566	+0.0039
	1.00	0.0512	0.589	+0.0025
	1.50†	0.0487	0.634	-
	2.00	0.0475	0.660	−0.0012
	3.00	0.0458	0.693	−0.0029
	4.00 (best)	0.0450	0.706	−0.0037
	JSD span		**0.0077**	

The monotonic JSD reduction with larger T_{dec} reflects the role of T_{dec} in undoing RLHF logit compression: higher values widen the effective token-probability landscape so that IS perturbations exert larger directional influence. However, r peaks near $T_{\text{dec}}=1.0$ and declines at 2.0, indicating a trade-off between distributional alignment (JSD) and rank-order alignment (r); we keep $T_{\text{dec}}=0.5$ as a deliberately conservative operating point.

F Dataset Preprocessing and Sensitivity

Pipeline. Each MultiTP CSV is processed in five steps: (i) deduplication keeps only the canonical paraphrase (`which_paraphrase = 0`); (ii) the Utilitarianism *quality filter* drops scenarios where the two groups have identical size *and* all characters belong to the *quality-attribute* role set {PREGNANT, WOMAN, LARGEWOMAN}, which would otherwise inject a non-numerosity signal into a dimension whose AMCE is by construction the count-difference effect; (iii) per-category capping retains at most 80 scenarios per dimension; (iv) under-represented categories are oversampled

with replacement to a minimum of 36 scenarios; and (v) the resulting pool is shuffled with a fixed seed (42).

Reproducible left/right assignment. When the MultiTP `paraphrase_choice` field does not unambiguously determine which group is rendered on the left vs. right, we deterministically fall back to a SHA-256 hash of the tuple (sub_1, sub_2, g_1, g_2) modulo 2, so the same scenario receives the same ordering across runs and machines. The fallback rate is logged per country; for all 15 countries it stays below 5%.

Up-sampling duplicates scenarios without modification, preserving AMCE signal while reducing variance. Category capping prevents dimension dominance. Side balancing enables effective debiasing. To evaluate whether these choices alter the effective evaluation distribution, we compare *six* preprocessing configurations on 10 representative countries (USA, DEU, CHN, JPN, BRA, VNM, GBR, KOR, RUS, NGA) using Llama-3.1-70B.

Table 8: Per-country dataset sensitivity. JSD range = 0.0034 (aggregate) and < 0.010 in 9/10 countries; the only exception is RUS at 0.013, driven by augmentation sensitivity in the rare Species/Utilitarianism categories. D0 is retained as default.

Country	D0-Default	D1-NoAug	D2-Cap40	D3-Cap120	D4-NoFlip	D5-Strict
USA	0.0655	0.0674	0.0625	0.0644	0.0643	0.0684
DEU	0.0465	0.0515	0.0492	0.0467	0.0444	0.0484
CHN	0.0393	0.0363	0.0385	0.0383	0.0409	0.0398
JPN	0.0304	0.0291	0.0319	0.0296	0.0273	0.0320
BRA	0.0492	0.0551	0.0479	0.0510	0.0520	0.0480
VNM	0.0592	0.0574	0.0631	0.0585	0.0578	0.0626
GBR	0.0614	0.0667	0.0608	0.0618	0.0597	0.0633
KOR	0.0371	0.0354	0.0378	0.0357	0.0349	0.0382
RUS	0.0383	0.0489	0.0389	0.0374	0.0383	0.0357
NGA	0.0522	0.0568	0.0506	0.0525	0.0514	0.0538
Mean	**0.0479**	0.0505	0.0481	0.0476	0.0471	0.0490
Δ vs D0	-	$+0.0025$	$+0.0002$	-0.0003	-0.0008	$+0.0011$

D1-NoAug shows the largest degradation ($+0.0025$), justifying synthetic augmentation of the under-represented Species and Utilitarianism categories. D4-NoFlip is marginally lower than D0, indicating that group-side randomisation is conservative (it adds noise) rather than harmful; we retain it because it preserves positional-debiasing integrity. The 0.0034 aggregate range is well below the ± 0.004 bootstrap CI, so reported results are not confounded by preprocessing choices.

G Held-Out τ Calibration

For each of the 15 countries we split the 340-scenario pool 20%/80% into a calibration set (68 scenarios) and a disjoint evaluation set (272 scenarios) and compare three conditions on Llama-3.1-70B: **(A)** τ calibrated on the held-out 20% slice and evaluated on the 80% slice (correct protocol; this is the configuration used in all main-paper experiments); **(B)** τ calibrated on the evaluation set itself (simulated leakage); and **(C)** fixed τ=0.001 (no calibration).

The aggregate gap $|\text{JSD}_B - \text{JSD}_A| = 0.0006$ is smaller than the ± 0.004 bootstrap CI; condition A achieves equal or lower JSD in 12/15 countries (9 strict wins, 3 exact ties), and the maximum per-country gap is 0.0041 (BRA), still within noise. Notably $\tau_A < \tau_B$ in 12/15 countries (mean ratio $1.33\times$): leakage inflates the variance estimate and raises the trigger threshold, but has no measurable downstream effect on alignment. This robustness is theoretically expected because τ observes a single scalar (inter-agent logit variance) with no access to AMCE labels and targets a fixed 35% trigger rate rather than any alignment objective. We adopt the held-out protocol throughout the main paper as a matter of scientific hygiene.

H Cross-Lingual Token Elicitation

Native-language prompting system. Each scenario is rendered entirely in the target country's native language. The implementation includes full translations of: (i) scenario framing and context sentences (4 paraphrase variants per language), (ii) all 22 character types with singular/plural forms

Table 9: Per-country held-out τ calibration. $|\Delta\text{JSD}|$ remains below the bootstrap CI (± 0.004) in all 15 countries; held-out τ is strictly more conservative (mean $\tau_B/\tau_A = 1.33$) yet alignment is unchanged.

Country	τ_A	τ_B	τ_B/τ_A	JSD_A	JSD_B	$\lvert\Delta\text{JSD}\rvert$	$A \leq B$
USA	0.332	0.586	1.77	0.0620	0.0637	0.0017	✓
DEU	0.809	0.496	0.61	0.0377	0.0387	0.0010	✓
CHN	0.250	0.285	1.14	0.0344	0.0346	0.0002	✓
JPN	0.824	1.109	1.35	0.0335	0.0334	0.0001	≈
BRA	1.563	1.961	1.25	0.0606	0.0647	0.0041	✓
SAU	0.773	1.305	1.69	0.0533	0.0534	0.0001	✓
VNM	0.131	0.247	1.89	0.0599	0.0600	0.0001	✓
GBR	0.289	0.411	1.42	0.0636	0.0646	0.0010	✓
KOR	0.210	0.196	0.93	0.0504	0.0504	0.0000	≈
RUS	0.091	0.110	1.21	0.0442	0.0442	0.0000	≈
MEX	0.355	0.520	1.46	0.0383	0.0387	0.0004	✓
NGA	0.922	0.891	0.97	0.0495	0.0491	0.0004	≈
AUS	0.285	0.389	1.37	0.0690	0.0704	0.0014	✓
FRA	0.906	1.555	1.72	0.0486	0.0485	0.0001	≈
IND	0.371	0.461	1.24	0.0479	0.0479	0.0000	≈
Mean	0.541	0.701	1.33	**0.0502**	0.0508	0.0007	12/15

in each language, (iii) lane labels and group identifiers in native script, and (iv) closing questions with language-appropriate conjunctions and grammar.

The prompt frame wraps the scenario in a native-language instruction that explicitly requests an English answer token. Each frame follows the pattern: "[moral dilemma preamble in native language] \n {scenario} \n [instruction to answer A or B in English]". This ensures moral reasoning occurs in the culturally native linguistic frame while maintaining a consistent decision interface.

Token verification. Token IDs are verified per model at initialisation by encoding the upper-case strings `"A"` and `"B"` with `add_special_tokens=False` and asserting single-token mappings: Qwen2.5: A=22397, B=11572; Llama-3.1: A=31266, B=9268. Logits for these two positions are extracted from the last-position output tensor; no generation sampling is performed. The model uses its built-in `chat_template` for correct role formatting across model families.

I Baseline Implementation Details

I.1 Activation Steering

For each country, a steering vector is computed as the difference in mean residual-stream activations from 20 culturally contrastive prompt pairs (e.g., "Answer as someone from [country] who values [WVS-high trait]" vs. "Answer as someone with the opposite values"). Vectors are extracted from layer 32 (transformer midpoint) of Llama-3.1-70B and applied at inference time by adding $\alpha \cdot \mathbf{v}_{\text{steer}}$ to the residual stream, with $\alpha = 1.5$ selected via the same synthetic validation set used for SWA-PTIS.

Sensitivity analysis and fairness considerations. Activation steering performance is known to be sensitive to (i) the extraction layer, (ii) the scaling coefficient α, and (iii) the construction of contrastive pairs [Arditi et al., 2025]. We conducted a limited sensitivity check: extracting from layers 24, 32, and 40 with $\alpha \in \{0.5, 1.0, 1.5, 2.0, 3.0\}$, we found that layer 32 with $\alpha = 1.5$ yielded the lowest mean JSD (0.0877); other configurations ranged from 0.089 to 0.112. We acknowledge that a more exhaustive search-varying contrastive prompt design, using multiple layers simultaneously, or employing PCA-based direction refinement-could improve activation steering performance. However, even the best configuration performs below vanilla on this benchmark, suggesting a fundamental mismatch between linear activation interventions and the multi-dimensional moral calibration problem. The key structural advantage of SWA-PTIS is that it operates *per-scenario* with adaptive correction magnitude, whereas activation steering applies a fixed direction uniformly across all scenarios.

I.2 ARGS and Controlled Decoding: from Incompatible to Reformulated

ARGS [Khanov et al., 2024] and controlled decoding [Mudgal et al., 2024] were originally formulated with trained reward models, which in the cross-cultural setting would demand a separate reward

model per country-unavailable for MultiTP. However, the binary forced-choice structure of MultiTP admits a natural reformulation in which the reward is a closed-form functional of the persona logit gaps $\{\delta_i\}$. We exploit this in Appendix O (**Binary-ARGS**) to construct three reward specifications (ARGS-UNIF, ARGS-WVS, ARGS-PT) that share SWA-PTIS's batched forward pass and differ only in the reward kernel. The strongest variant, ARGS-PT, recovers the SWA-PTIS kernel exactly, providing an independent route to the same optimum.

J Detailed Ablation Analysis

Always-on vs. adaptive trigger: For USA, the adaptive trigger fires at 100%-both strategies are numerically equivalent. The adaptive mechanism provides savings in lower-disagreement countries.

Utilitarian persona: Removing it collapses Utilitarianism MPR from 61.2% to 48.0% (human: 76.6%), confirming it anchors the numerosity dimension.

Debiasing: Mean positional bias is 0.193 logit units; Species preference inflates from 85.8% to 93.5% without it (human: 79.2%).

K Trigger Mechanism and Latency

The adaptive τ targets 35% activation but achieves 100% for USA (all scenarios contested). The *flip rate* (IS reversals) is 6.0%, indicating confidence modulation rather than decision reversal.

Mean latency: 730ms (SWA-PTIS) vs. 160ms (vanilla) on Qwen2.5-72B (H100). Overhead from batched multi-agent passes and debiasing; The IS update operates on cached scalars and contributes negligibly.

L Germany Pearson r Analysis

Germany's JSD remains unchanged (0.0603) while Pearson r drops from 0.845 to 0.747. In 6D, Pearson r is sensitive to rank perturbations: a single dimension swap reduces r by $\sim$0.1 while JSD (depending on magnitudes, not ordering) stays constant. Rare IS corrections introduce minor rank perturbations without affecting distributional distance.

M Per-Country Results for All Models

Tables 9–13 present the full per-country breakdown (vanilla and SWA-PTIS absolute JSD, Pearson r, and percentage improvement) for all five evaluated models. The main-text Table **??** aggregates the percentage column across the four operating-regime models for compactness; the absolute numbers below are the source of those percentages.

N Consensus-Shift Baselines

To isolate the contribution of PT-IS's importance-weighted sampling from the simpler benefit of using persona consensus at all, we implement four deterministic shift baselines applied directly to the scalar consensus gap $\bar{\delta}$, all sharing the same batched forward pass and two-pass debiasing as SWA-PTIS: CS-MEAN ($\delta_{\text{opt}} = \bar{\delta}$, equivalent to B5 PersonaConsensus), CS-STD ($\bar{\delta} \pm \sigma_{\text{agents}}$ in the consensus direction), CS-SCALED ($\bar{\delta}(1 + \alpha\,\text{Var})$), and CS-CLAMP ($\text{clip}(1.5\bar{\delta}, \pm 1)$).

CS-MEAN reproduces B5 PersonaConsensus exactly (cross-validating our ablation hierarchy). CS-STD *degrades* relative to CS-MEAN in 9/15 countries-variance amplification without the asymmetric loss-sensitive utility actively hurts in high-disagreement regimes. CS-CLAMP is the strongest deterministic shift; its largest improvements occur in high inter-persona-variance countries (USA -0.015, AUS -0.016, GBR -0.012), exactly where Prospect Theory's asymmetric weighting is most active. The residual aggregate gap CS-CLAMP$\rightarrow$SWA-PTIS (ΔJSD ≈ 0.003) is attributed to PT's gain/loss asymmetry, which no scalar shift can replicate.

Table 10: Per-country results for **Qwen2.5-72B-Instruct**.

Country	Lang.	JSD ↓		Pearson r ↑		JSD Improv.	
		Van.	SWA	Van.	SWA	(Δ)	(%)
USA	en	.1066	**.0308**	0.400	0.822	−.076	+71.1
Germany	de	.0603	.0603	0.845	0.747	+.000	+0.0
China	zh	.0513	.0371	0.851	0.762	−.014	+27.8
Japan	ja	.0986	.0553	0.634	0.786	−.043	+43.9
Brazil	pt	.0754	.0703	0.349	0.529	−.005	+6.7
Saudi Arabia	ar	.0778	.0796	0.443	0.527	+.002	−2.3
Vietnam	vi	.1302	.0783	0.569	0.456	−.052	+39.9
France	fr	.0789	.0479	0.617	0.838	−.031	+39.4
India	hi	.0600	.0561	0.670	0.739	−.004	+6.5
South Korea	ko	.0720	.0513	0.355	0.645	−.021	+28.8
Great Britain	en	.1057	**.0294**	0.419	0.800	−.076	+72.2
Russia	ru	.0800	.0629	0.623	0.784	−.017	+21.4
Mexico	es	.0760	.0534	0.605	0.545	−.023	+29.8
Nigeria	en	.1103	.0649	0.407	0.615	−.046	+41.2
Australia	en	.1047	**.0258**	0.438	0.841	−.079	+75.4
Mean		**.0859**	**.0536**	0.548	0.696	−.032	**+37.6**

Table 11: Per-country results for **Llama-3.1-70B-Instruct**.

Country	Lang.	JSD ↓		Pearson r ↑		JSD Improv.	
		Van.	SWA	Van.	SWA	(Δ)	(%)
USA	en	.1561	.1234	0.012	0.197	−.033	+20.9
Germany	de	.0709	.0391	−0.455	0.685	−.032	+44.9
China	zh	.0459	.0311	0.406	0.802	−.015	+32.3
Japan	ja	.0457	.0416	0.496	0.466	−.004	+9.0
Brazil	pt	.0532	.0656	−0.370	0.348	+.012	−23.3
Saudi Arabia	ar	.0539	.0596	0.043	−0.210	+.006	−10.7
Vietnam	vi	.1546	.0529	−0.519	−0.135	−.102	+65.8
France	fr	.0500	.0516	0.250	0.394	+.002	−3.0
India	hi	.0442	.0337	−0.315	0.715	−.010	+23.6
South Korea	ko	.0893	.0436	−0.272	0.551	−.046	+51.2
Great Britain	en	.1568	.0918	0.012	0.286	−.065	+41.5
Russia	ru	.0449	.0413	0.539	0.637	−.004	+7.9
Mexico	es	.1265	.0425	−0.371	0.307	−.084	+66.4
Nigeria	en	.1582	.0988	0.109	0.497	−.059	+37.5
Australia	en	.1551	.1010	0.039	0.224	−.054	+34.9
Mean		**.0937**	**.0612**	−0.026	0.384	−.033	**+34.7**

O Binary-ARGS Reward-Guided Decoding

We adapt ARGS [Khanov et al., 2024] to the binary forced-choice setting as *Binary-ARGS*: K perturbations $\delta_k = \bar{\delta} + \epsilon_k$ ($\epsilon \sim \mathcal{N}(0, \sigma^2)$) are sampled and re-weighted by $\exp(r_k/\beta)$, where r_k is one of three closed-form rewards. ARGS-UNIF uses a uniform directional reward ($\text{sign}(\bar{\delta}) \cdot \delta_k/N$), equivalent to importance-weighted consensus averaging; ARGS-WVS weights each persona by its WVS cultural salience; ARGS-PT replaces the linear reward by the Prospect-Theory utility identical to the SWA-PTIS kernel.

Three findings emerge. **(1)** ARGS-UNIF (JSD 0.0524) closely matches B5 PersonaConsensus, confirming uniform importance weighting reduces to consensus averaging. **(2)** ARGS-WVS (JSD 0.0544) is *worse* than ARGS-UNIF in 12/15 countries: WVS salience weighting without asymmetric loss-sensitivity actively misaligns persona contributions. **(3)** ARGS-PT (JSD 0.0485, r 0.514) is best/tied in 13/15 countries and dominates on every aggregate metric, empirically showing that SWA-PTIS is precisely an ARGS variant whose reward is a Prospect-Theory utility over a Gaussian proposal. The two countries where ARGS-PT underperforms (BRA, IND) have the highest inter-persona variance and overcorrect in the absence of the distributional $\text{KL}(q \parallel p)$ trust region that

Table 12: Per-country results for **Qwen2.5-7B-Instruct** (4-bit).

Country	Lang.	JSD ↓		Pearson r		JSD Improv.	
		Van.	SWA	Van.	SWA	(Δ)	(%)
USA	en	.0531	.0448	0.048	0.531	−.008	+15.5
Germany	de	.0530	.0522	0.207	0.212	−.001	+1.4
China	zh	.0512	.0455	0.401	0.397	−.006	+11.2
Japan	ja	.0527	.0540	0.490	0.449	+.001	−2.3
Brazil	pt	.0445	.0331	0.415	0.837	−.011	+25.8
Saudi Arabia	ar	.0539	.0443	0.234	0.265	−.010	+17.9
Vietnam	vi	.0547	.0472	0.259	0.263	−.008	+13.6
France	fr	.0529	.0442	0.259	0.315	−.009	+16.5
India	hi	.0570	.0524	0.466	0.183	−.005	+8.1
South Korea	ko	.0492	.0329	0.239	0.363	−.016	+33.2
Great Britain	en	.0520	.0302	0.140	0.483	−.022	+41.9
Russia	ru	.0506	.0452	0.373	0.491	−.005	+10.5
Mexico	es	.0479	.0384	0.038	0.360	−.010	+19.9
Nigeria	en	.0551	.0411	0.101	0.425	−.014	+25.3
Australia	en	.0550	.0431	0.170	0.452	−.012	+21.7
Mean		**.0522**	**.0426**	0.252	0.402	−.010	**+18.4**

Table 13: Per-country results for **Llama-3.1-8B-Instruct** (4-bit).

Country	Lang.	JSD ↓		Pearson r		JSD Improv.	
		Van.	SWA	Van.	SWA	(Δ)	(%)
USA	en	.0470	.0333	−0.323	0.498	−.014	+29.1
Germany	de	.0716	.0529	−0.101	0.382	−.019	+26.1
China	zh	.0561	.0457	−0.082	0.388	−.010	+18.5
Japan	ja	.0518	.0519	0.545	0.395	+.000	−0.2
Brazil	pt	.0732	.0628	−0.222	0.109	−.010	+14.2
Saudi Arabia	ar	.0551	.0513	−0.252	−0.230	−.004	+6.9
Vietnam	vi	.0505	.0423	−0.330	0.303	−.008	+16.2
France	fr	.0515	.0612	−0.610	−0.220	+.010	−18.8
India	hi	.0756	.0564	−0.471	0.347	−.019	+25.3
South Korea	ko	.0522	.0419	−0.352	−0.062	−.010	+19.7
Great Britain	en	.0375	.0255	−0.065	0.457	−.012	+32.0
Russia	ru	.0560	.0434	−0.230	0.593	−.013	+22.5
Mexico	es	.0439	.0383	−0.079	0.344	−.006	+12.7
Nigeria	en	.0474	.0410	−0.229	0.303	−.006	+13.5
Australia	en	.0505	.0363	−0.156	0.328	−.014	+28.1
Mean		**.0533**	**.0436**	−0.247	0.271	−.011	**+22.6**

SWA-PTIS's σ_0-floored Gaussian proposal enforces at the sampler level (see Appendix C.2 for the distributional-KL interpretation of the softmax update).

P Entropy-Aware Adaptive σ for Qwen2.5-32B

We study whether an entropy-aware adaptive σ can address the Qwen2.5-32B failure mode (Section 5). At each scenario we measure the Shannon entropy H of the top-50 token distribution at the decision position and set $\sigma = \sigma_{\text{base}} \cdot \text{clip}(H/H_{\text{ref}},\ 0.2,\ 5.0)$ with H_{ref}=2.5 nats and σ_{base}=0.30. We compare against a fixed-σ sweep on 8 countries (USA, DEU, CHN, JPN, BRA, VNM, GBR, KOR).

The mean entropy is $H = 0.521$ nats, far below H_{ref}, confirming concentrated logits. Adaptive σ self-scales to $\bar{\sigma} = 0.086$ ($r(H, \sigma)$=0.948), validating the diagnostic but collapsing onto the low-σ regime. The best fixed σ=0.80 reaches JSD 0.0599 vs. 0.0613 at default-a marginal absolute gain-while the per-country breakdown exposes the deeper problem: countries cluster into a high-σ group (USA/JPN/GBR/KOR) and a low-σ group (DEU/CHN/BRA/VNM), but *all* countries share $H \approx 0.52$ nats, so a per-scenario entropy signal cannot discriminate them. A complete fix therefore requires *jointly* adaptive τ per country to detect which countries have sufficient logit variance to benefit from large perturbations-left as future work.

Table 14: Per-country results for **Qwen2.5-32B-Instruct** (4-bit). SWA-PTIS degrades in 7/15 countries due to concentrated logit distributions.

Country	Lang.	JSD ↓		Pearson r		JSD Improv.	
		Van.	SWA	Van.	SWA	(Δ)	(%)
USA	en	.0447	.0440	0.887	0.738	−.001	+1.5
Germany	de	.0587	.0534	0.893	0.803	−.005	+9.1
China	zh	.0507	.0574	0.827	0.818	+.007	−13.1
Japan	ja	.0485	.0592	0.846	0.805	+.011	−22.0
Brazil	pt	.0717	.0645	0.552	0.422	−.007	+9.7
Saudi Arabia	ar	.0883	.1015	0.704	0.625	+.013	−14.9
Vietnam	vi	.0900	.0780	0.634	0.556	−.012	+13.1
France	fr	.0730	.0681	0.647	0.652	−.005	+6.7
India	hi	.0737	.0840	0.458	0.583	+.010	−13.8
South Korea	ko	.0529	.0303	0.106	0.064	−.023	+43.0
Great Britain	en	.0470	.0345	0.889	0.732	−.013	+26.5
Russia	ru	.0700	.0780	0.721	0.711	+.008	−11.3
Mexico	es	.0775	.0933	0.632	0.381	+.016	−20.3
Nigeria	en	.0641	.0512	0.124	0.156	−.013	+20.1
Australia	en	.0541	.0425	0.883	0.781	−.012	+21.4
Mean		**.0632**	**.0633**	0.625	0.550	+.000	**−0.2**

Table 15: Per-country JSD for the four consensus-shift baselines on Llama-3.1-70B (15 countries). CS-CLAMP is the strongest deterministic shift but still loses to SWA-PTIS in aggregate; CS-STD is catastrophic for NGA (+0.017 vs. CS-MEAN) due to unbounded variance amplification.

Country	CS-Mean	CS-Std	CS-Scaled	CS-Clamp
USA	0.0923	0.0919	0.0933	**0.0771**
DEU	**0.0410**	0.0448	0.0449	0.0428
CHN	0.0591	0.0618	0.0608	**0.0520**
JPN	**0.0306**	0.0335	0.0343	0.0311
BRA	0.0524	0.0519	0.0530	**0.0459**
SAU	0.0565	0.0584	0.0570	**0.0536**
VNM	0.0632	0.0659	0.0669	**0.0564**
FRA	0.0547	0.0516	0.0552	**0.0442**
IND	0.0421	**0.0410**	0.0413	0.0418
KOR	0.0451	0.0454	0.0451	**0.0443**
GBR	0.0800	0.0842	0.0844	**0.0681**
RUS	0.0450	**0.0435**	0.0443	0.0429
MEX	0.0539	0.0539	0.0531	**0.0472**
NGA	0.0826	0.0998	0.0987	**0.0799**
AUS	0.0868	0.0864	0.0861	**0.0711**
Mean	0.0590	0.0609	0.0612	**0.0532**
SWA-PTIS	**0.0502** (Pearson r = 0.424)			

Q Extended Country Pipeline Run (Engineering Check)

We run the SWA-PTIS pipeline end-to-end on 8 additional countries not in the MultiTP benchmark, spanning 4 script systems and 5 language families: Poland (pl), Turkey (tr), Indonesia (id), Argentina (es), Ukraine (uk), Taiwan (zh), the Philippines (tl), and Egypt (ar). **This appendix makes an engineering claim, not a scientific one.** Because no Moral Machine human AMCEs are available for these countries, we evaluate on 300 synthetic scenarios per country whose Vanilla AMCE is uniformly 50.0 by construction. The only thing this experiment establishes is that the persona-construction, translation, batched-forward-pass, τ-calibration, and IS-update code runs to completion across the 4 scripts and 5 language families above. The $\sim$+0.05pp "AMCE shift" is therefore measuring perturbation activity, not alignment, and the mean JSD numbers should not be compared to the main table. We retain the appendix only as evidence that the framework is not silently English-specific.

Because the synthetic scenarios uniformly initialise Vanilla AMCE at 50.0, the shift metric measures only the pipeline's ability to modulate output distributions away from an uninformative prior; it cannot be interpreted as alignment improvement over human preferences and is not directly comparable to the main 15-country results. We include this probe purely to demonstrate that the WVS-grounded

Table 16: Binary-ARGS per-country results (Llama-3.1-70B, 15 countries). ARGS-PT achieves best or tied JSD in 13/15 countries and best r in 11/15, confirming SWA-PTIS is precisely an ARGS variant with a Prospect-Theory utility over a Gaussian proposal.

	JSD ↓			Pearson r ↑		
Country	Unif	WVS	PT	Unif	WVS	PT
USA	0.0757	0.0793	**0.0646**	0.506	0.484	**0.626**
DEU	0.0476	0.0512	**0.0465**	0.668	0.656	**0.732**
CHN	0.0347	0.0396	0.0362	0.753	0.721	**0.788**
JPN	0.0308	0.0311	**0.0275**	0.782	0.757	**0.831**
BRA	**0.0482**	0.0509	0.0519	0.351	0.317	**0.375**
SAU	0.0541	0.0539	0.0534	0.076	0.079	**0.165**
VNM	0.0717	0.0755	**0.0551**	−0.507	−0.427	−0.369
GBR	0.0692	0.0709	**0.0588**	0.494	0.488	**0.616**
KOR	0.0421	0.0386	0.0364	0.722	**0.750**	0.748
RUS	**0.0399**	0.0443	0.0415	0.723	0.728	**0.782**
MEX	0.0464	0.0496	**0.0428**	0.255	0.256	**0.503**
NGA	0.0567	0.0598	**0.0502**	0.775	0.779	**0.852**
AUS	0.0742	0.0763	**0.0664**	0.461	0.452	**0.557**
FRA	0.0495	0.0484	0.0480	0.336	0.370	**0.423**
IND	**0.0457**	0.0463	0.0475	0.045	−0.003	**0.079**
Mean	0.0524	0.0544	**0.0485**	0.429	0.427	**0.514**

Table 17: Per-country JSD under fixed-σ on Qwen2.5-32B. Two latent country clusters emerge: USA/JPN/GBR/KOR strongly prefer high σ (USA: $0.0523 \rightarrow 0.0317$ at σ=1.0); DEU/CHN/BRA/VNM strictly prefer low σ. All 8 countries share $H \approx 0.52$ nats, so the entropy signal alone cannot discriminate them.

σ	USA	DEU	CHN	JPN	BRA	VNM	GBR	KOR	Mean
0.05	0.0515	0.0647	0.0533	0.0690	0.0692	0.0710	0.0500	0.0670	0.0620
0.10	0.0519	0.0648	0.0532	0.0684	0.0694	0.0711	0.0499	0.0668	0.0619
0.15	0.0522	0.0649	0.0530	0.0676	0.0698	0.0711	0.0499	0.0665	0.0619
0.20	0.0518	0.0651	0.0527	0.0661	0.0702	0.0713	0.0498	0.0661	0.0616
$0.30^{\dagger}$	0.0523	0.0654	0.0516	0.0644	0.0710	0.0717	0.0491	0.0649	0.0613
0.50	0.0487	0.0661	0.0515	0.0601	0.0732	0.0734	0.0476	0.0604	0.0601
0.80	**0.0427**	0.0672	0.0599	**0.0585**	0.0780	0.0757	**0.0439**	**0.0532**	**0.0599**
1.00	0.0317	0.0681	0.0714	0.0637	0.0833	0.0744	0.0458	0.0522	0.0613
Adaptive ($\bar{\sigma}$=0.086)	0.0516	0.0647	0.0532	0.0688	0.0693	0.0711	0.0500	0.0670	0.0620

persona construction, the batched forward pass, the two-pass debiasing, and the IS loop all run end-to-end across right-to-left Arabic, CJK, Cyrillic, and several Latin-script languages without any code change. Future work should collect human AMCE annotations for these countries to enable full quantitative evaluation.

R Extended Limitations

Single seed and stochasticity. All reported numbers use one run per model with one IS random draw (K=128 perturbations) per scenario; variance across seeds is not characterised here.

Pearson r in six dimensions. As noted in Section 5, the AMCE vector is short, so per-country r has large sampling error and small row-wise r deltas are not necessarily statistically separable; JSD remains the primary claim-bearing metric.

WVS-to-trolley linkage. The mapping from WVS value dimensions to trolley moral dimensions follows the indirect-modulation construction in Appendix B.3, not a fully validated causal chain; we therefore avoid claims of the form "WVS dimension X caused AMCE shift Y."

Inter-persona reward assumption. The reward $r_i = \delta_i - \delta_{\text{base}}$ assumes persona shifts approximate human target directions. This is plausible when the base model represents a WEIRD-biased prior, but if a persona's shift is orthogonal to the human target, the signal may mislead. Per-dimension validation is needed.

Table 18: Extended country pipeline run (Llama-3.1-70B). The pipeline executes end-to-end for 4 script systems and 5 language families; IS-induced AMCE shifts are detected in all 8 countries (mean $+0.05 \pm 0.02$ pp).

Country	Lang.	Script	AMCE Shift	Vanilla JSD	SWA JSD
Poland (POL)	Polish	Latin	+0.03 pp	0.0496	0.0497
Turkey (TUR)	Turkish	Latin	+0.04 pp	0.0376	0.0377
Indonesia (IDN)	Indonesian	Latin	+0.06 pp	0.0425	0.0424
Argentina (ARG)	Spanish	Latin	+0.04 pp	0.0388	0.0387
Ukraine (UKR)	Ukrainian	Cyrillic	+0.07 pp	0.0485	0.0483
Taiwan (TWN)	Chinese	CJK	+0.03 pp	0.0427	0.0426
Philippines (PHL)	Filipino	Latin	+0.06 pp	0.0396	0.0396
Egypt (EGY)	Arabic	Arabic	+0.06 pp	0.0511	0.0513
Mean			$\mathbf{+0.05 \pm 0.02}$ **pp**	0.0438	0.0438

Geographic coverage, preprocessing, and survey vintage. We evaluate 15 of 100+ MultiTP countries; results depend on the preprocessing pipeline in Appendix F. WVS Wave 7 ($\sim$2017–2022) is a fixed historical slice; persona quality is weakest where coverage is sparse (e.g., Saudi Arabia, Brazil), as reflected in per-country tables. No low-resource languages (Amharic, Khmer, Yoruba) appear in the panel; cross-lingual generalisation to such settings remains untested.

Binary format. The logit-difference formulation requires generalisation to softmax for multi-option settings.

Language–culture conflation. One language per country may conflate linguistic and cultural effects.

Model quantisation. 4-bit/Int8 quantisation may introduce logit artefacts affecting IS stability.

WVS temporal staleness. Wave 7 ($\sim$2017–2022) data may not reflect current preferences in rapidly evolving societies.

Qwen2.5-32B failure mode. Concentrated logit distributions (H=0.521 nats) cause IS perturbations to land off-manifold and yield near-uniform importance weights. Appendix P shows that an entropy-aware σ correctly diagnoses the cause but cannot fix it on its own, because all 8 evaluated countries share the same low entropy and thus cannot be discriminated by a per-scenario entropy signal. A complete remedy requires *jointly* adaptive (σ, τ) per country and is left to future work.